Notes: Abaluck & Gruber (AER, 2011)

Choice Inconsistencies among the Elderly: Evidence from Plan Choice in the Medicare Part D Program

Contents

1 Big picture

- **Question.** Do elderly consumers choose Medicare Part D prescription-drug plans as if they fully understand the financial trade-offs across premiums, expected out-of-pocket costs, risk, and plan characteristics?
- **Setting.** The paper studies the 2006 launch year of Medicare Part D, where elderly consumers faced dozens of stand-alone prescription drug plans that differed in premiums, formularies, deductibles, donut-hole coverage, cost sharing, and quality.
- **Core data advantage.** The authors match individual prescription-drug claims from Wolters Kluwer to the CMS universe of Part D plan characteristics, allowing them to simulate each person's total costs under every plan in the choice set.
- **Main descriptive fact.** Only 12.2% of consumers choose the lowest-cost plan under perfect foresight. On average, consumers could reduce total Part D spending by about 30.9% by switching to the lowest-cost plan.
- **Main behavioral result.** Consumers overweight premiums relative to expected out-of-pocket costs, place large value on plan characteristics beyond their direct cost/risk consequences, and appear to undervalue variance reduction.
- **Robustness result.** The premium/OOP-cost inconsistency and the direct valuation of plan characteristics survive many robustness checks, private-information corrections, and random-coefficients models. The variance result is more sensitive.
- **Welfare result.** Partial-equilibrium welfare would be substantially higher if consumers made plan choices consistent with the rational benchmark; the abstract reports roughly 27% higher welfare under rational choices.
- **Policy takeaway.** More plan choice is valuable only if consumers can evaluate plans well. Choice architecture, plan simplification, defaults, and decision aids are central to the success of privatized public insurance.

2 Incremental contribution of the paper

1. **From plan choice to individual cost simulation.** The paper does not simply compare premiums or plan popularity. It simulates each individual's expected financial consequences under every plan in the actual county-level choice set.
2. **Direct evidence on choice inconsistencies in a high-stakes public insurance market.** The setting is not a lab experiment or a small employer plan. It is a nationwide public benefit delivered through private plan choice.
3. **Separates multiple margins of poor choice.** The paper distinguishes premium/OOP misweighting, plan-feature overvaluation, and variance underweighting rather than treating poor choice as a single error.
4. **Links plan complexity to welfare.** The paper quantifies how much welfare is lost when choices deviate from a rational cost-risk benchmark.
5. **Addresses private information and heterogeneity.** The authors explicitly consider that consumers may know more about future drug needs than the econometrician and that preferences may be heterogeneous.
6. **Policy contribution.** The paper provides early evidence that Medicare Part D and similar insurance exchanges need not only many options, but also tools that help consumers compare economically relevant features.

Interpretation: The incremental contribution is to make the quality of insurance choice measurable. By comparing every chosen plan to the same person's full choice set, the paper converts plan complexity into a quantitative measure of foregone savings and welfare loss.

3 Data and measurement

3.1 Medicare Part D setting

Medicare Part D began in 2006 and offered prescription-drug coverage through private insurers. Stand-alone prescription drug plans differed on premiums, deductibles, formularies, quality, cost sharing, generic coverage, and donut-hole coverage. A typical consumer faced more than 40 stand-alone plans.

Interpretation: The setting is ideal for testing choice quality because consumers face a rich menu with meaningful financial consequences and substantial variation in plan attributes.

3.2 Claims data and plan matching

The main data come from Wolters Kluwer prescription drug transactions for people over age 65 in 2005 and 2006. The authors match claims to CMS plan characteristics. The final conditional-logit sample used in Table 1 has 95,742 patients, 702 plans, 47 states, and 36 brands.

Interpretation: The key data innovation is that the authors can simulate what the same consumer would have paid in every available plan, rather than observing only the chosen plan.

3.3 Expected cost and risk measures

For each individual i and plan j, the paper constructs:

- premium $Premium_j$;
- out-of-pocket cost OOP_{ij};
- variance of out-of-pocket costs $\mathrm{Var}(\mathrm{OOP}_{ij})$;
- plan design measures: deductible, donut-hole coverage, generic coverage, cost sharing, formulary breadth, and quality.

The benchmark financial cost is:

$$TotalCost_{ij} = Premium_j + E[\mathrm{OOP}_{ij}].$$

Interpretation: The total-cost measure is the core object behind the foregone-savings calculations. The variance term is included to allow for risk aversion rather than assuming consumers only minimize expected spending.

3.4 Foregone savings

For each individual, foregone savings are:

$$ForegoneSavings_i = TotalCost_{i,chosen} - \min_{j \in J_i} TotalCost_{ij}.$$

Interpretation: A large value does not automatically prove irrationality because consumers may value risk protection, quality, convenience, or private information. The structural models are designed to test whether these rationalizations can explain the observed choices.

4 Models and empirical specifications

4.1 Expected utility benchmark

The paper begins with a constant absolute risk aversion setup. If costs are normally distributed,

$$U(C) = -\exp[-\gamma(W - C)], \qquad C \sim N(\mu, \sigma^2).$$

A risk-averse consumer acts as if she minimizes a certainty-equivalent cost:

$$CE_{ij} = Premium_j + E[\text{OOP}_{ij}] + \frac{\gamma}{2}\text{Var}(\text{OOP}_{ij}).$$

Interpretation: This benchmark clarifies what a rational plan chooser should value: lower premiums, lower expected OOP costs, and lower variance. Other plan characteristics should matter only to the extent that they affect utility through costs, risk, quality, or convenience.

4.2 Conditional logit model

The main empirical choice model is a conditional logit over plans:

$$U_{ij} = \beta_p Premium_j + \beta_o E[\text{OOP}_{ij}] + \beta_v \text{Var}(\text{OOP}_{ij}) + \delta' X_j + \alpha_{brand} + \varepsilon_{ij}.$$

Here X_j includes financial plan characteristics such as deductible, donut-hole coverage, generic coverage, cost sharing, formulary breadth, and quality.

Interpretation: If consumers optimize only over their own expected costs and risk, premiums and expected OOP costs should have similar dollar weights. If $|\beta_p|$ is much larger than $|\beta_o|$, consumers overweight premiums relative to OOP costs.

4.3 Choice inconsistencies

The paper tests three inconsistencies:

1. **Premium/OOP inconsistency:** premiums receive far more weight than out-of-pocket costs.
2. **Plan-characteristic inconsistency:** features such as donut-hole coverage, deductible, and generic coverage receive large direct utility weights beyond their effects on the individual's simulated costs and risk.
3. **Variance inconsistency:** variance-reducing plan features are valued weakly, implying low or unstable estimates of risk aversion.

Interpretation: The first two are the most robust. The variance result depends more heavily on measurement of risk and assumptions about private information.

4.4 Private information model

To allow consumers to know more than the econometrician about future costs, the paper models a fraction of cost variation as private information. Table 3 estimates the percent private information and adjusts the variance term accordingly.

Interpretation: Private information can explain some differences between realized and expected costs, but it does not eliminate the core finding that premiums receive much greater weight than OOP costs.

4.5 Random-coefficients model

The random-coefficients model allows heterogeneous preferences over premiums, expected OOP costs, variance, quality, and plan characteristics.

$$\beta_i = \bar{\beta} + \nu_i, \qquad \nu_i \sim N(0, \Sigma_\beta).$$

Interpretation: If the conditional-logit results were an artifact of ignoring taste heterogeneity or IIA, random coefficients should change the mean coefficients substantially. Table 4 shows that the qualitative conclusions persist.

5 Identification and empirical design

5.1 What is identified

The paper identifies how observed plan choices trade off premiums, expected OOP costs, risk, and plan attributes. It is not a randomized policy experiment. Identification comes from cross-sectional variation in plan characteristics within each consumer's choice set and from the ability to simulate each consumer's cost under each plan.

Interpretation: The choice model asks: conditional on the plans available to a consumer and the predicted costs of those plans, which plan features predict the actual selected plan?

5.2 Key identifying variation

- Different counties and states have different plan menus.
- Plans within a county differ in premiums, deductibles, formularies, donut-hole coverage, generic coverage, quality, and cost-sharing designs.
- The same plan characteristic can have different cost implications for different individuals because drug utilization differs.
- The model compares chosen and unchosen plans within an individual choice set.

Interpretation: The within-choice-set comparison is important because it controls for the fact that different consumers face different menus. The econometrician asks why the person chose plan j rather than plan k from the same menu.

5.3 Main threats and responses

1. **Expected OOP measurement error.** Realized costs are noisy proxies for expected costs. The paper uses robustness checks and private-information models.
2. **Private information.** Consumers may know future drug needs better than claims-based predictions. Table 3 estimates and allows for substantial private information.
3. **Unobserved heterogeneity.** Consumers may value plan attributes differently. Table 4 uses random coefficients.
4. **Plan quality and brands.** Quality, brand dummies, and brand-state dummies absorb some nonfinancial plan differences.
5. **Dual eligibles and matching error.** Robustness checks restrict samples and alter matching rules.

Interpretation: The identification is strongest for the premium/OOP-cost gap because it survives many alternative assumptions. The variance result is more tentative because the variance measure is especially sensitive to private information and risk assumptions.

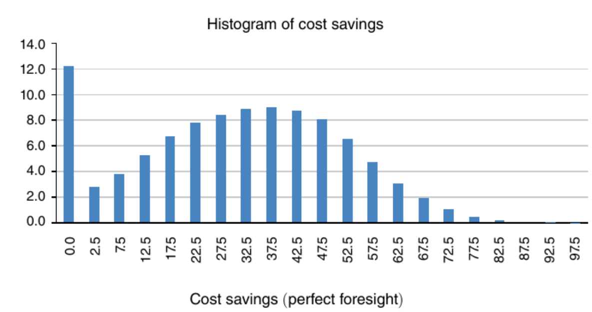
6 Tables and figures

6.1 Figure 1: Histogram of Cost Savings from Switching to Lowest-Cost Plan; Figure 2: Average Mean and Standard Deviation for Each PDP Plan in California

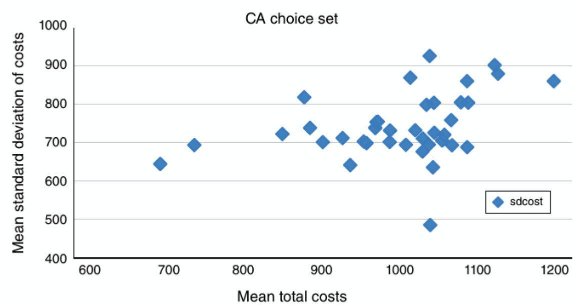
Read: Figure 1 shows the distribution of potential cost savings from switching to the lowest-cost plan. Figure 2 plots the average mean and standard deviation of costs for PDP plans in California.

- Only 12.2% choose the lowest-cost plan.
- The average consumer could save 30.9% of total Part D spending by switching to the lowest-cost plan.
- Many plans are far from the efficient frontier in mean-variance space.

Economic message: The descriptive evidence shows that plan choice errors are economically large. Figure 2 also weakens the idea that consumers simply buy higher mean costs to reduce risk; many high-cost plans do not offer proportionately lower risk.



(a) Figure 1: Cost savings from switching to the lowest-cost plan.



(b) Figure 2: California plan mean-variance choice set.

Cost minimization and mean-variance evidence. Figure 1 documents large foregone savings; Figure 2 shows that many plans are not attractive mean-variance trade-offs.

6.2 Figure 3: Percent Choosing Donut Hole Coverage and Added Cost by Expenditure Quantile

Read: Figure 3 compares the share choosing donut-hole coverage with the added cost of such coverage by expenditure quantile.

- The share choosing donut-hole coverage is almost flat across the spending distribution at around 10%.
- The financial value of donut-hole coverage differs dramatically by expenditure quantile.
- Low-expenditure consumers choose donut-hole coverage about as often as high-expenditure consumers, despite having much less to gain from it.

Economic message: Consumers appear to value the label or feature of donut-hole coverage directly rather than pricing its individualized financial value.

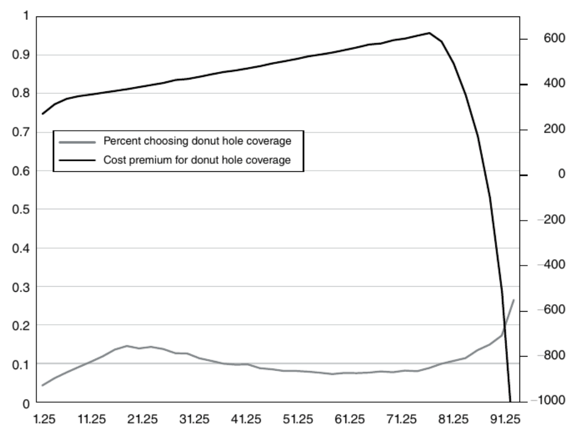


FIGURE 3. PERCENT CHOOSING DONUT HOLE COVERAGE AND ADDED COST BY EXPENDITURE QUANTI

Figure 3: Percent Choosing Donut Hole Coverage and Added Cost by Expenditure Quantile. The flat choice curve contrasts with the highly variable financial value of coverage, motivating the plan-characteristic inconsistency tests.

TABLE 1—CONDITIONAL LOGIT RESULTS

	(1)	(2)	(3)	(4)
Premium	−0.4330***	−0.7663***	−0.4990***	−0.5218***
(hundreds)	(0.0029)	(0.0038)	(0.0061)	(0.0069)
OOP costs (realized)	−0.2127***	−0.1172***	−0.0961***	−0.0967***
(hundreds)	(0.1520)	(0.0015)	(0.0015)	(0.0016)
Variance	−0.0189***	−0.0004	−0.0006	−0.0005
(times 10^6)	(0.0027)	(0.0007)	(0.0007)	(0.0007)
Deductible	x	−0.2899***	−0.1628***	−0.1674***
(hundreds)		(0.0049)	(0.0067)	(0.0072)
Donut hole	x	3.023***	1.762***	1.865***
		(0.0181)	(0.0277)	(0.0303)
Generic coverage	x	0.4203***	0.3004***	0.2700***
		(0.0140)	(0.0175)	(0.0177)
Cost sharing	x	3.282***	1.189***	1.057***
		(0.0538)	(0.0741)	(0.0778)
Number of top 100 on form	x	0.0937***	0.0587***	0.0644***
		(0.0007)	(0.0017)	(0.0018)
Average quality	0.4091***	0.7398***	x	x
	(0.0032)	(0.0039)		
Brand dummies	No	No	Yes	No
Brand-state dummies	No	No	No	Yes
Risk index	10	0	0	0
Number of patients	95,742	95,742	95,742	95,742
Number of plans	702	702	702	702
Number of states	47	47	47	47
Number of brands	36	36	36	36

Notes: The table shows conditional logit results from estimating the model given in equation (6) by maximum likelihood. Each column shows coefficients from a single regression. The coefficients reported are the parameters of the utility function, not marginal effects. Standard errors are in parentheses. The first column includes only premium, realized out-of-pocket costs, and the variance measure. The second column adds controls for the indicated plan characteristics, the third column adds brand fixed effects, and the fourth column adds brand-state fixed effects. Premiums, out-of-pocket costs, and deductibles are in hundreds of dollars and the variance term is in millions. The cost-sharing variable is computed as the average value of covered expenditures divided by total drug expenditures for individuals in the choice set. The average quality variable is a normalized version of the "average rating" index provided by CMS. The risk index is twice the coefficient on the variance divided by the coefficient on premiums scaled by 100. In the model in the text, this value equals one million times the coefficient of absolute risk aversion.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

Table 1: Conditional Logit Results. The table shows that premiums receive much more weight than expected OOP costs and that plan features matter directly.

6.3 Table 1: Conditional Logit Results

Read: Table 1 is the core conditional-logit table.

- Premium coefficients are much larger in magnitude than realized OOP-cost coefficients.
- In models with brand controls, the premium coefficient is more than five times the OOP-cost coefficient.
- Donut-hole coverage, deductible, generic coverage, cost sharing, and formulary breadth all enter directly and significantly.
- The variance coefficient is small and often near zero once richer plan features are included.

Economic message: Consumers act as if a dollar of premium matters much more than a dollar of expected OOP costs. They also value plan features independently of their predicted financial consequences.

Interpretation: Table 1 is the main evidence of choice inconsistency. The key reading is not any single coefficient, but the relative dollar weights on premiums and OOP costs plus the direct valuation of plan-design features.

6.4 Table 2: Robustness Checks

Read: Table 2 reestimates the conditional-logit model under alternative matching rules, samples, claim restrictions, and quality-variable definitions.

- The premium/OOP-cost gap remains across robustness checks.
- Deductible and donut-hole coefficients remain economically large.
- Results survive broader and narrower plan-match definitions and different treatment of low-claim individuals.

Economic message: The core results are not driven by a single sample construction rule or matching algorithm.

TABLE 2—ROBUSTNESS CHECKS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Premium	–0.4099***	–0.5717***	–0.5099***	–0.4935***	–0.5517***	–0.5220***	–0.7422***
(hundreds)	(0.0070)	(0.0076)	(0.0074)	(0.0072)	(0.0078)	(0.0070)	(0.0043)
OOP costs (realized)	–0.0508***	–0.0808***	–0.0846***	–0.0977***	–0.0986***	–0.0944***	–0.1100***
(hundreds)	(0.0016)	(0.0018)	(0.0018)	(0.0016)	(0.0016)	(0.0015)	(0.0015)
Variance	–0.0009	–0.0067***	–0.0011	–0.0003	0.0002	–0.0007	–0.0004
(times 10^6)	(0.0010)	(0.0029)	(0.0014)	(0.0007)	(0.0012)	(0.0008)	(0.0008)
Deductible	–0.0456***	–0.2680***	–0.2179***	0.4378***	0.4876***	–0.1723***	–0.2730***
(hundreds)	(0.0072)	(0.0070)	(0.0068)	(0.0095)	(0.0100)	(0.0072)	(0.0051)
Donut hole	1.224***	2.023***	1.907***	2.872***	3.956***	1.837***	2.935***
	(0.0302)	(0.0299)	(0.0296)	(0.0346)	(0.0380)	(0.0304)	(0.0204)
Generic coverage	–0.0241	0.1168***	0.1345***	0.7570***	0.8362***	0.2597***	0.4206***
	(0.0158)	(0.0173)	(0.0296)	(0.0233)	(0.0265)	(0.0178)	(0.0141)
Full cost sharing	0.9973***	2.129***	1.319***	1.357***	–0.0986	1.076***	3.633***
	(0.1313)	(0.1316)	(0.1295)	(0.0838)	(0.0897)	(0.0781)	(0.0555)
Number of top 100 on form	0.0737***	0.0664***	0.0599***	0.1173***	0.1126***	0.0654***	0.0952***
	(0.0013)	(0.0015)	(0.0014)	(0.0023)	(0.0022)	(0.0018)	(0.0007)
Customer service	x	x	x	x	x	x	0.1644***
							(0.0066)
Prescription filling	x	x	x	x	x	x	0.5666***
							(0.0037)
Pricing avail./changes	x	x	x	x	x	x	0.4142***
							(0.0054)
Missing quality dummies	No	No	No	No	No	No	Yes
Brand dummies	No	No	No	No	No	No	No
Brand-state dummies	Yes	Yes	Yes	Yes	Yes	Yes	No
Risk index	0	0	0	0	0	0	0
Number of patients	99,999	99,981	99,981	99,988	99,999	99,994	95,742
Number of plans	888	861	861	702	702	702	702
Number of states	49	49	49	47	47	47	47
Number of brands	48	42	42	36	36	36	36

Notes: Table shows conditional logit results from estimating the model given in equation (6) by maximum likelihood on different subsamples to check robustness. Each column shows coefficients from a single regression. The coefficients reported are the parameters of the utility function, not marginal effects. Standard errors are in parentheses. Premiums, out-of-pocket costs and deductibles are in hundreds of dollars and the variance term is in millions. The cost-sharing variable is computed as the average value of covered expenditures divided by total drug expenditures for individuals in the choice set. The first column estimates the model on a random sample of 100,000 patients selected from the sample of individuals who were matched to a PDP plan in the "Lax match" discussed in the text. The second column estimates the model on the original sample used in Table 1 plus individuals with zero claims in 2005. The third column uses the same sample as the second column, but dropping individuals with zero claims in 2005, and more than 12 claims in 2006. The fourth column includes only unique matches and multiple matches which could be assigned with 95 percent certainty. The fifth column includes only unique matches. The sixth column uses the original sample but restricting to individuals for which more than 50 percent of their claims were non-dual. The seventh column disaggregates the quality variable. Note that the seventh column also does not include brand or brand-state dummies, since these are collinear with the quality variables. The risk index is twice the coefficient on the variance divided by the coefficient on premiums scaled by 100. In the model in the text, this value equals one million times the coefficient of absolute risk aversion.

***Significant at the 1 percent level.

Table 2: Robustness Checks. The premium/OOP cost gap and plan-feature effects remain under alternative sample and matching assumptions.

6.5 Table 3: Results with Private Information; Table 4: Random Coefficients Results

Read: Table 3 allows consumers to possess private information about future costs; Table 4 allows heterogeneous preferences through random coefficients.

- Table 3 estimates that consumers know about 58% of the variation in future costs not predictable by the econometrician's cells.
- Even after allowing private information, premiums remain far more heavily weighted than OOP costs.
- Table 4 shows that random coefficients do not eliminate the mean preference pattern.
- Heterogeneity is statistically important, but average coefficients remain close to the baseline.

Economic message: Private information and heterogeneity matter, but they do not rationalize the main premium/OOP-cost inconsistency or the direct valuation of plan features.

Table 3—Results with Private Information

	Final sample	Restricted sample	Private info
Percent private information			0.5818***
			(0.0618)
Premium	−0.7383***	−0.7156***	−0.7489***
(hundreds)	(0.0038)	(0.0094)	(0.0132)
OOP costs (realized)	−0.1169***	−0.1040***	−0.1687***
(hundreds)	(0.0016)	(0.0039)	(0.0094)
Variance	−0.0026	−0.1103**	−0.8574***
(times 10^6)	(0.0014)	(0.0517)	(0.2947)
Deductible	−0.2677***	−0.3079***	−0.2767***
(hundreds)	(0.0014)	(0.1257)	(0.0137)
Donut hole	2.823***	2.805***	2.870***
	(0.0181)	(0.0490)	(0.0478)
Generic coverage	0.3066***	0.4743***	0.4784***
	(0.0143)	(0.0341)	(0.0347)
Full cost sharing	2.990***	3.391***	2.829***
	(0.0546)	(0.1417)	(0.1743)
Number of top 100 on form	0.0939***	0.0995***	0.1005***
	(0.0007)	(0.0019)	(0.0021)
Average quality	0.7167***	0.7418***	0.7512***
	(0.0039)	(0.0098)	(0.0095)
Brand dummies	No	No	No
Brand-state dummies	No	No	No
Risk index	1	31	229
Number of patients	95,742	15,001	15,001
Number of plans	702	702	702
Number of states	47	47	47
Number of brands	36	36	36

Notes: The table compares conditional logit results with results from estimating the random coefficients model given in equations (11) and (14) using the Laplace approximation to the likelihood function developed by Matthew C. Harding and Jerry A. Hausman (2007) with bootstrapped standard errors. Each column shows coefficients from a single regression. The coefficients reported are the parameters of the utility function, not marginal effects. Standard errors are in parentheses. The first column is identical to the second column of Table 1. The second column estimates the same model on a random subsample of 15,000, and the third column estimates the random coefficients model on this same subsample. Variable definitions are identical to Table 1. The "percent private information" field corresponds to the variable τ_{frac} in the model in the text.

***Significant at the 1 percent level.

(a) Table 3: Private information correction.

Table 4—Random Coefficients Results

	(1)	(2)	(3)
Premium	−0.7156***	−0.7677***	−0.7354***
(hundreds)	(0.0094)	(0.0111)	(0.0841)
Standard deviation of premium	x	0.2940***	0.2659***
		(0.0560)	(0.0330)
OOP costs (realized)	−0.1040***	−0.1091***	−0.1193***
(hundreds)	(0.0039)	(0.0052)	(0.0171)
Standard deviation of OOP costs	x	0.0001	0.0226
		(0.0003)	(0.0506)
Variance	−0.1103***	−0.1486	−0.0866
(times 10^6)	(0.0517)	(0.0926)	(0.1237)
Standard deviation of variance	x	0.2035***	0.2603
		(0.0685)	(0.1769)
Deductible	−0.3079***	−0.2524***	−0.2354
(hundreds)	(0.1257)	(0.0107)	(0.1903)
Standard deviation of deductible	x	x	0.2922***
			(0.1452)
Donut hole	2.805***	2.775***	2.523***
	(0.0490)	(0.0630)	(0.3538)
Std. deviation of donut hole	x	x	0.8078
			(0.5997)
Generic coverage	0.4743***	0.4845***	−0.0037
	(0.0341)	(0.0427)	(0.2320)
Standard deviation of generic coverage	x		1.106***
			(0.5108)
Full cost sharing	3.391***	2.083***	1.829
	(0.1417)	(0.1852)	(1.304)
Standard deviation of cost share	x		.1290
			(.4828)
Number of top 100 on form	0.0995***	0.0992***	0.1405***
	(0.0019)	(0.0023)	(0.0078)
Standard deviation of top 100	x		0.1071
			(0.0790)
Average quality	0.7418***	0.7729***	0.7622***
	(0.0098)	(0.0090)	(0.0358)
Standard deviation of quality	x	0.3753***	0.3503
		(0.0146)	(0.2182)
Brand dummies	No	No	No
Brand-state dummies	No	No	No
Risk index	124	124	124
Number of patients	15,001	15,001	15,001
Number of plans	702	702	702
Number of states	47	47	47
Number of brands	36	36	36

Notes: Table shows results from estimating the random coefficients model discussed in the heterogeneity section, estimated using the Laplace approximation developed in Harding-Hausman (2007) with bootstrapped standard errors. This model is identical to the model in equation (6), adding the normally distributed noise term (which is a function of $C_{ij} - \mu_{ij}$) and

(b) Table 4: Random coefficients.

Private information and heterogeneity. These models address two major rationalizations for apparent mistakes. The main conclusions survive.

7 Short summary

- Medicare Part D offered elderly consumers large plan menus with financially meaningful differences.
- Consumers often failed to choose the lowest-cost plan.
- The majority of choices were off the efficient frontier of mean and variance.
- Consumers overweighted premiums relative to expected OOP costs.
- Consumers valued plan features, especially donut-hole coverage and deductibles, beyond their direct cost implications.
- Consumers placed weak or unstable value on variance reduction.
- The main inconsistencies survived robustness checks, private-information models, and random coefficients.
- The paper concludes that welfare would be much higher under rational plan choice.

8 Conclusion

8.1 Core conclusion

Abaluck and Gruber show that elderly consumers in Medicare Part D make plan choices that are difficult to reconcile with full-information expected-utility optimization. The most robust inconsistency is the enormous difference between how consumers respond to premiums and how they respond to expected out-of-pocket costs.

8.2 Economic intuition

Premiums are salient, fixed, and easy to compare. Out-of-pocket costs are uncertain, drug-specific, plan-specific, and difficult to forecast. Consumers therefore appear to use simple plan features and visible premiums as heuristics, even when those heuristics produce expensive choices.

8.3 Incremental contribution revisited

The paper's contribution is to quantify the cost of complexity in a major public insurance market. It shows that providing many private choices is not enough; consumers must also be able to map plan characteristics into expected financial consequences.

8.4 Policy implications

The results support stronger decision aids, plan standardization, smart defaults, active choice simplification, and regulatory attention to confusing plan features. The paper does not imply that consumers should have no choice, but it does show that unconstrained menu complexity can generate large welfare losses.

8.5 Limitations

The paper relies on simulated costs and must model expectations, private information, and heterogeneity. It cannot observe all dimensions of plan value such as service quality, pharmacy convenience, or subjective preferences. The variance result is more sensitive than the premium/OOP-cost and plan-feature results.

8.6 Takeaway

The paper's lasting message is that choice-based public insurance programs can fail when consumers cannot evaluate complex options. In such settings, market design matters as much as market availability.